



Data Communications and Networking

Fourth Edition

Chapter 13

Wired LANs: Ethernet



13-1 IEEE STANDARDS

In 1985, the Computer Society of the IEEE started a project, called Project 802, to set standards to enable intercommunication among equipment from a variety of manufacturers. Project 802 is a way of specifying functions of the physical layer and the data link layer of major LAN protocols.

Topics discussed in this section:

Data Link Layer

Physical Layer



IEEE Standards

❑ The data link layer has subdivided into two sublayers;

❖ Logical link control (LLC) and media access control (MAC)

LLC: Logical link control

MAC: Media access control

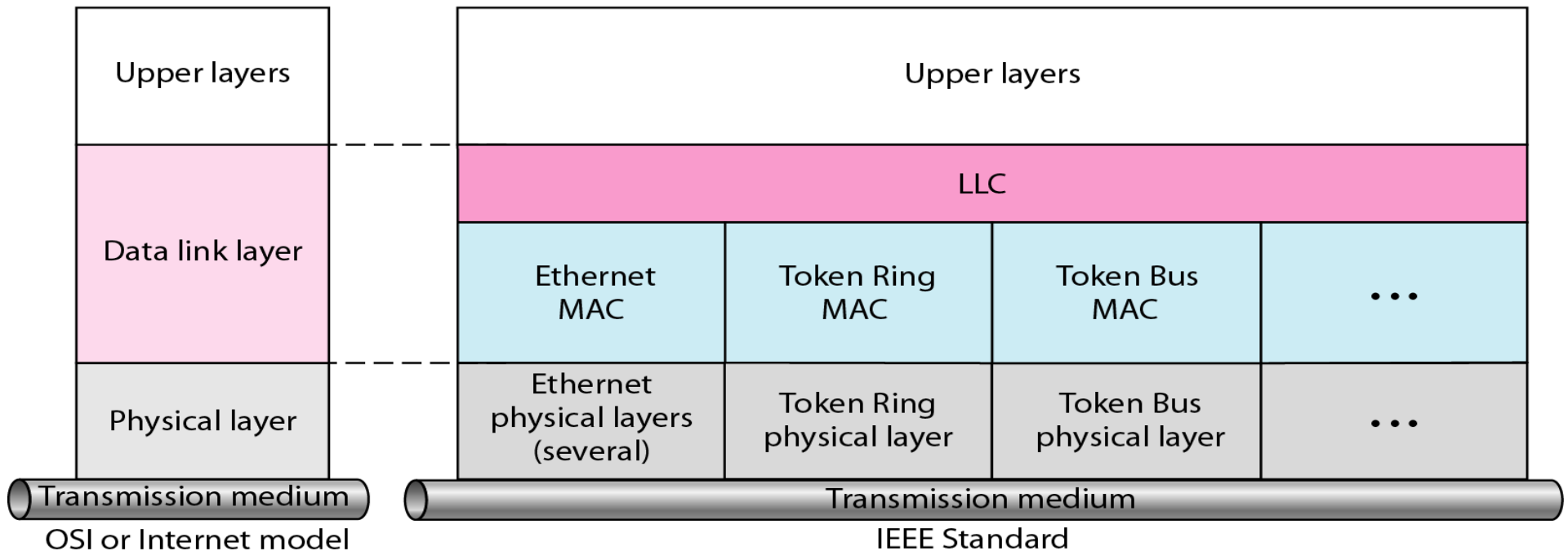


Figure 13.1 *IEEE standard for LANs*



IEEE Standards

❑ Logical Link Control (LLC)

❖ Flow control, error control, and part of the framing duties are collected into the LLC.

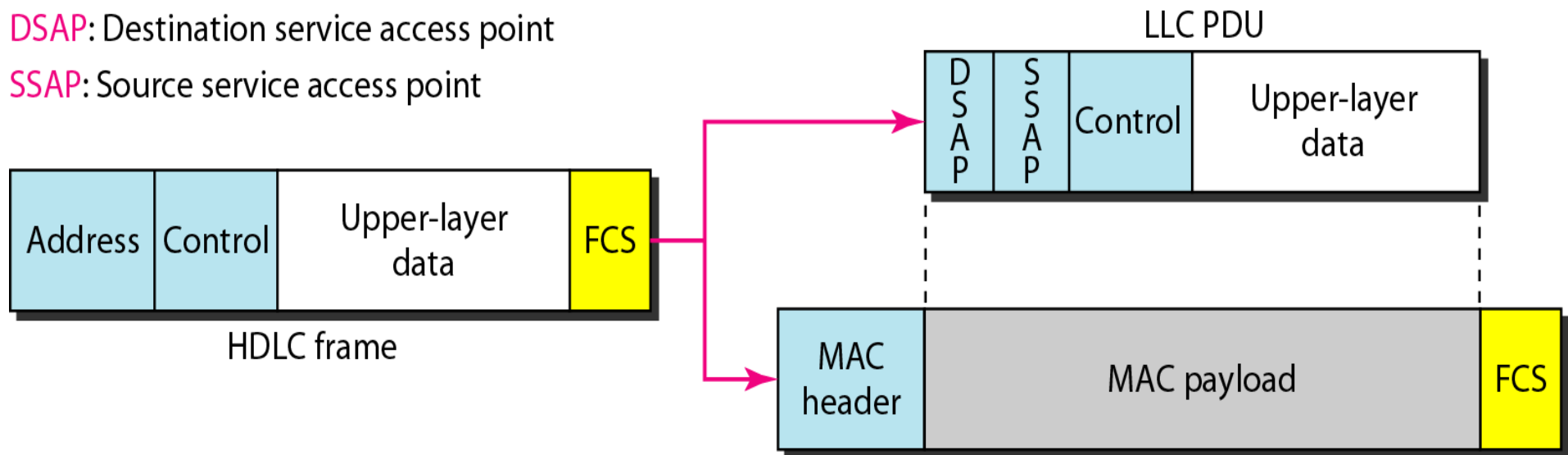
❑ Media Access Control (MAC)

❖ MAC defines the specific access method for each LAN.

❖ MAC contains a No. of distinct modules.

DSAP: Destination service access point

SSAP: Source service access point



13-2 STANDARD ETHERNET

*The original Ethernet was created in 1976 at Xerox's Palo Alto Research Center (PARC). Since then, it has gone through four generations. We briefly discuss the **Standard (or traditional) Ethernet** in this section.*

Topics discussed in this section:

MAC Sublayer
Physical Layer



Standard Ethernet

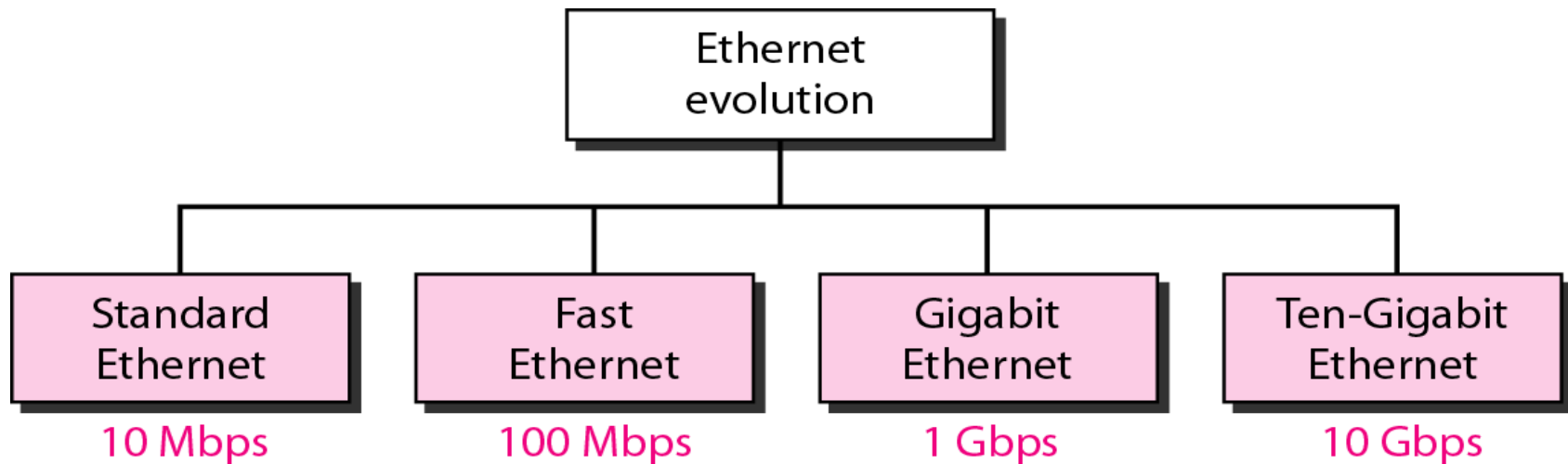


Figure 13.3 *Ethernet evolution through four generations*



MAC Sublayer

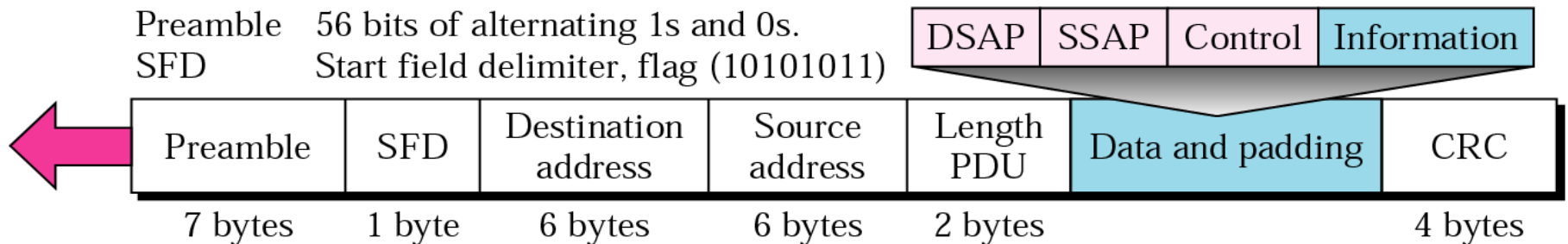


Figure 13.4 802.3 MAC frame

- Preamble – 7 bytes of alternating 0s and 1s to alert the receiver and allow it to synchronize
- Start Frame Delimiter (SFD) – 1 byte – 10101011 signals the beginning of a frame, last chance for synchronization – last 2 bits are 11
- Destination address (DA) – 6 bytes – contains the physical address of the destination station or stations
- Source address (SA) – 6 bytes – contains the physical address of the sender
- Length/type – it defines the length of the data field or to define upper-layer protocol using MAC frame
- Data – data encapsulated from upper-layer protocols : 46 ~ 1500 bytes
- CRC – CRC-32



MAC Sublayer

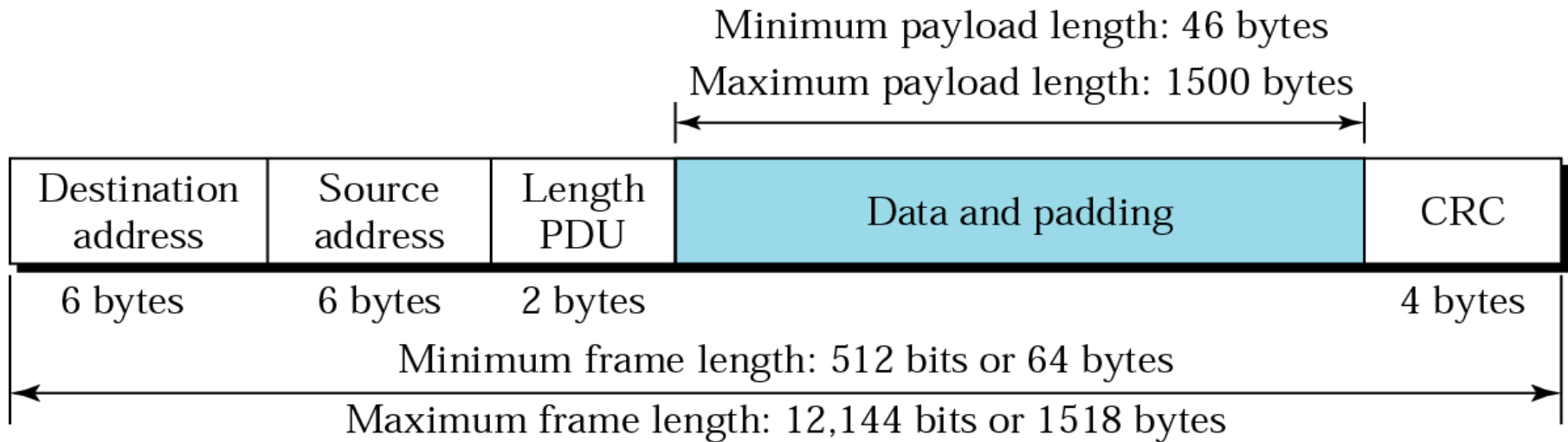


Figure 13.5 Minimum and maximum lengths

Frame length:

Minimum: 64 bytes (512 bits)

Maximum: 1518 bytes (12,144 bits)



MAC Addressing

- ❑ Ethernet addresses in hexadecimal notation

06 : 01 : 02 : 01 : 2C : 4B



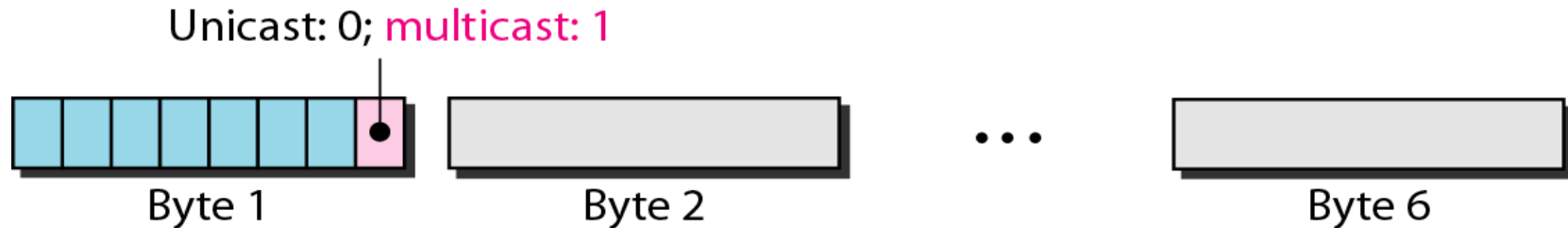
6 bytes = 12 hex digits = 48 bits

Figure 13.6 *Example of an Ethernet address in hexadecimal notation*

- ❑ Each station on an Ethernet network has its own network interface card (NIC)
- ❑ NIC provides the station with a 6-byte physical address



MAC Addressing



- ❖ The least significant bit of the first byte defines the type of address.
 - If the bit is **0**, the address is unicast; otherwise, it is multicast.
- ❖ The broadcast destination address is a special case of the multicast address in which all bits are 1s.

Figure 13.7 *Unicast and multicast addresses*

Access Method : CSMA/CD

- ❑ Standard Ethernet uses 1-persistent CSMA/CD

- ❑ Slot time

 - ❖ Slot time = round-trip time + time required to send the jam sequence (10Mbps Ethernet : 51.2μs)

- ❑ Slot time & Maximum Network Length

 - ❖ Max Length = Propagation speed x (slot time / 2)
$$= (2 \times 10^8) \times (51.2 \times 10^{-6} / 2) = 5,120\text{m}$$

 - ❖ Maximum Length of a traditional Ethernet network is reduced to 2,500m, because of the delay time in repeater and interfaces, and the time required to send the jam sequence.



Physical layer

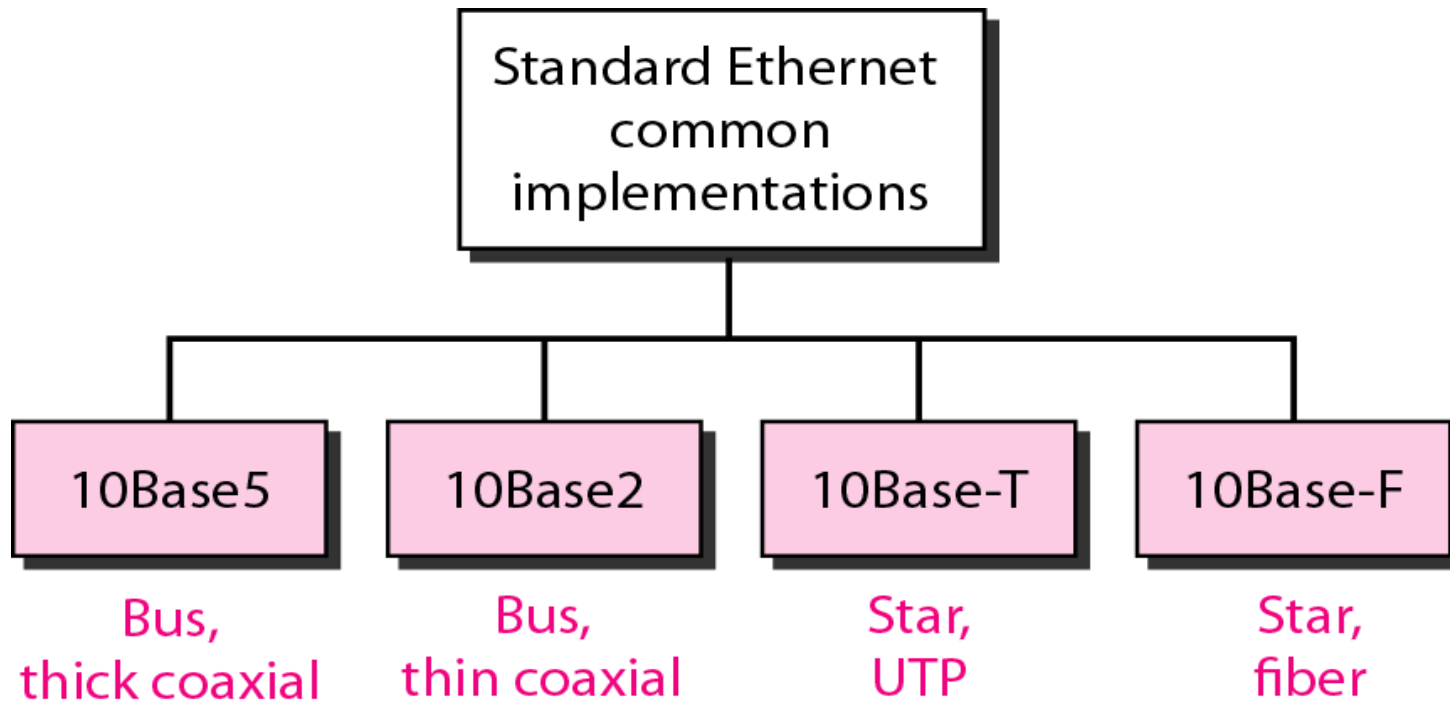


Figure 13.8 *Categories of Standard Ethernet*



Encoding and Decoding

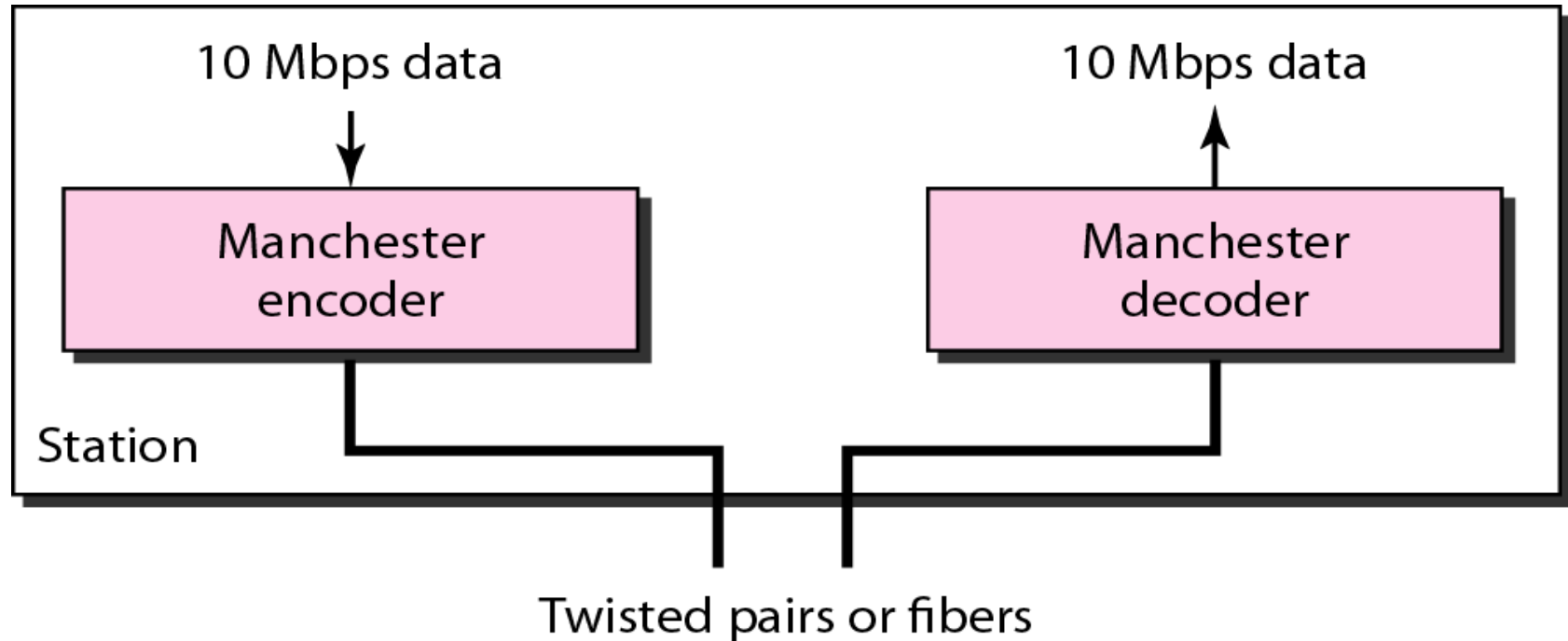


Figure 13.9 *Encoding in a Standard Ethernet implementation*

10 Base-5 : Thick Ethernet

- ❑ Thick Ethernet or Thicknet
- ❑ Bus topology, external transceiver
- ❑ Connection of a station to the medium using 10Base5

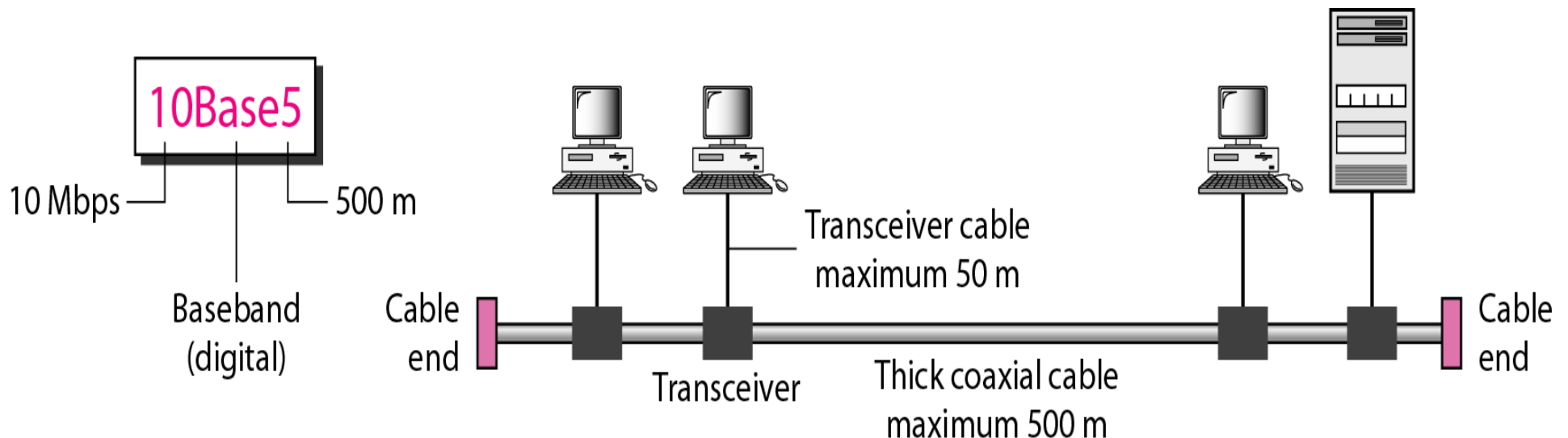


Figure 13.10 *10Base5 implementation*

10 Base-2 : Thin Ethernet

- ❑ Connection of stations to the medium using 10Base2
- ❑ Thin Ethernet or CheaperNet
- ❑ Bus topology, internal transceiver or a point-to-point connection via an external transceiver

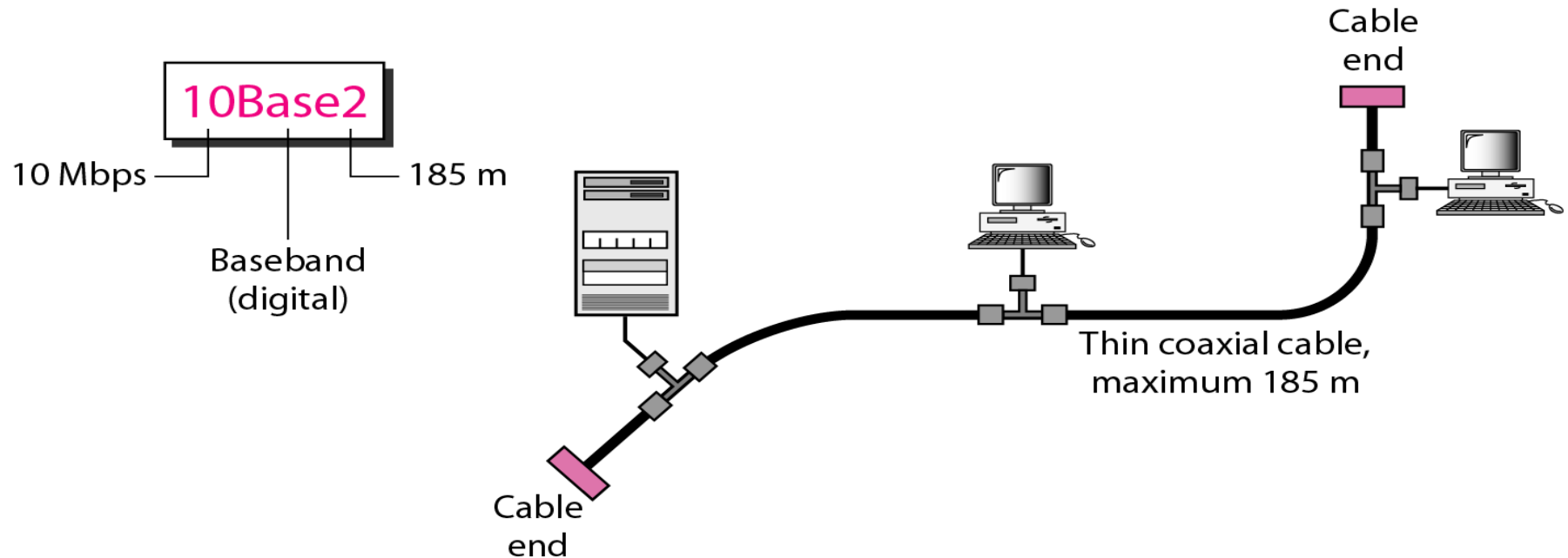


Figure 13.11 *10Base2 implementation*

10 Base-T : Twisted Pair Ethernet

- ❑ Physical star topology
- ❑ Stations connected to a hub with internal or external transceiver

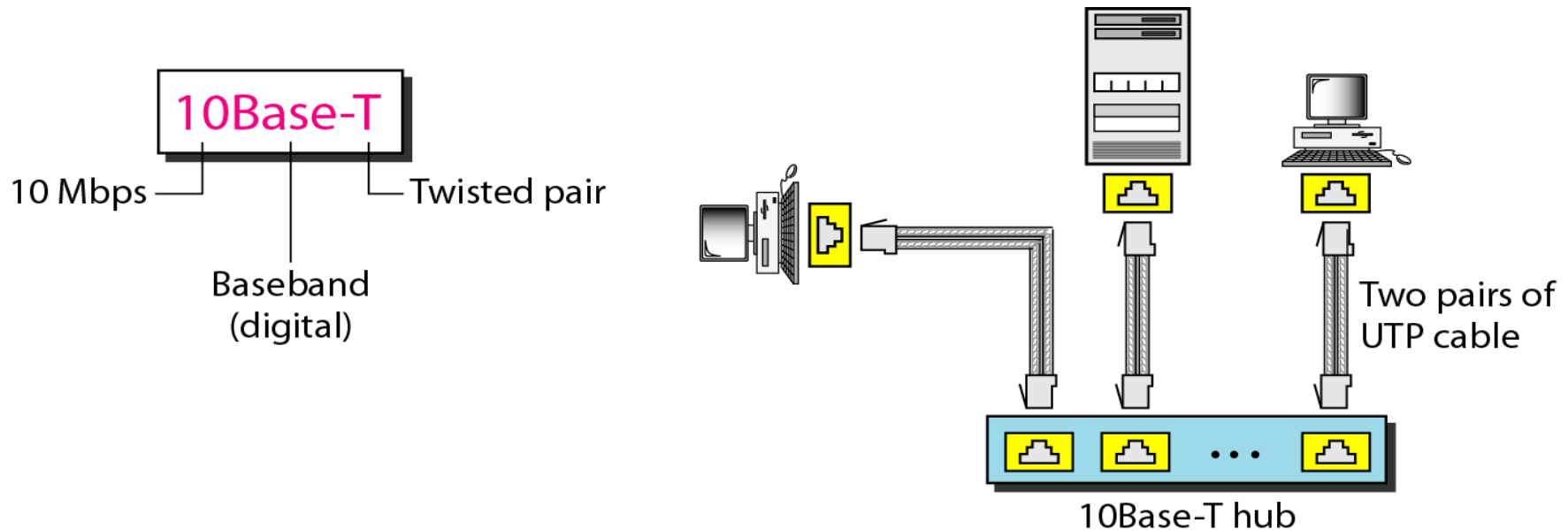


Figure 13.12 *10Base-T implementation*

10 Base-F : Fiber Ethernet

- ❑ Fiber link Ethernet
- ❑ Uses star topology to connect stations to a hub
- ❑ Normally implemented with external transceiver having two pairs of fiber-optic cables connecting it to the hub.

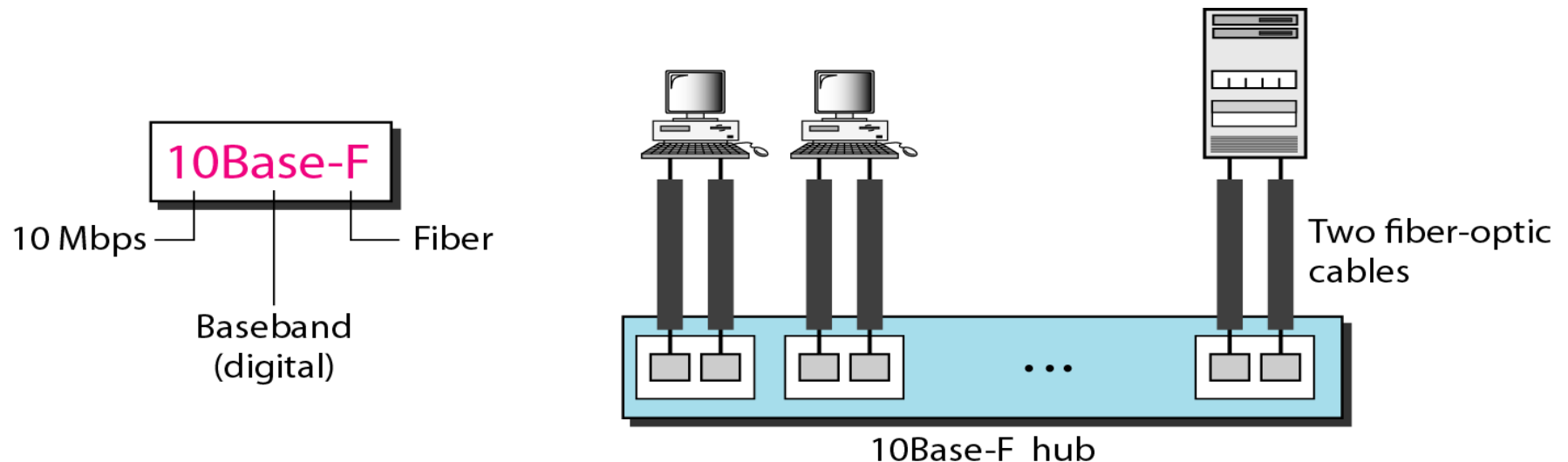


Figure 13.13 *10Base-F implementation*

Summary of Ethernet

Table 13.1 *Summary of Standard Ethernet implementations*

<i>Characteristics</i>	<i>10Base5</i>	<i>10Base2</i>	<i>10Base-T</i>	<i>10Base-F</i>
Media	Thick coaxial cable	Thin coaxial cable	2 UTP	2 Fiber
Maximum length	500 m	185 m	100 m	2000 m
Line encoding	Manchester	Manchester	Manchester	Manchester



13-3 CHANGES IN THE STANDARD

The 10-Mbps Standard Ethernet has gone through several changes before moving to the higher data rates. These changes actually opened the road to the evolution of the Ethernet to become compatible with other high-data-rate LANs.

Topics discussed in this section:

Bridged Ethernet

Switched Ethernet

Full-Duplex Ethernet



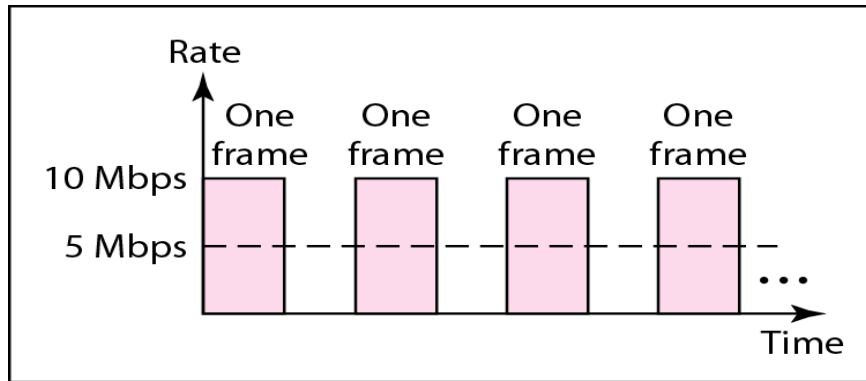
Bridged Ethernet

- ❑ Raising the bandwidth
- ❑ Separating collision domains

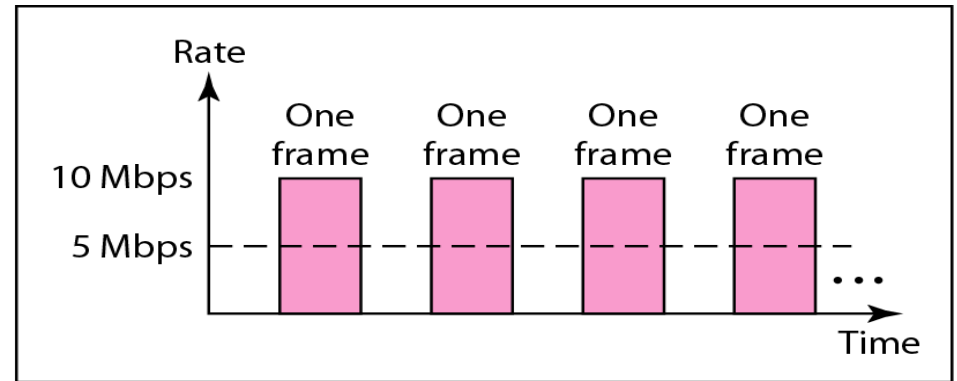


Raising the Bandwidth

Figure 13.14 *Sharing bandwidth*

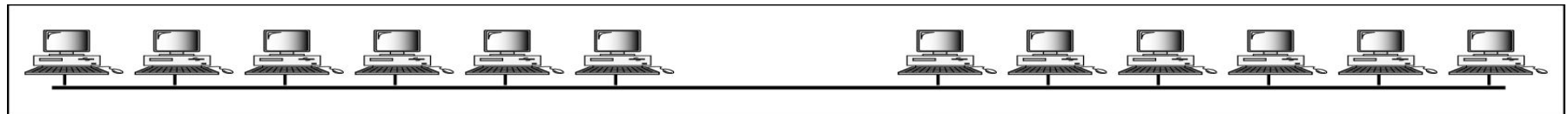


a. First station

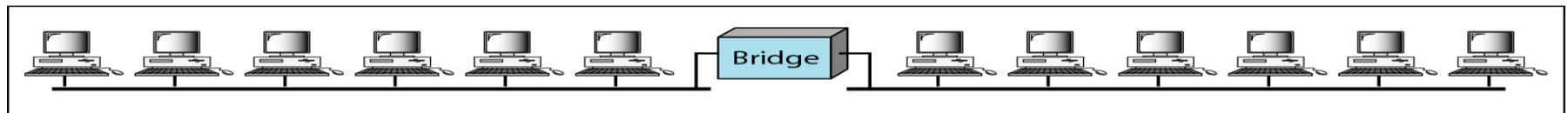


b. Second station

□ A Network with and without a Bridge



a. Without bridging



b. With bridging

❖ 10/6 Mbps vs 10/12 Mbps in case that traffic is not going through the bridge



Separating Collision Domains

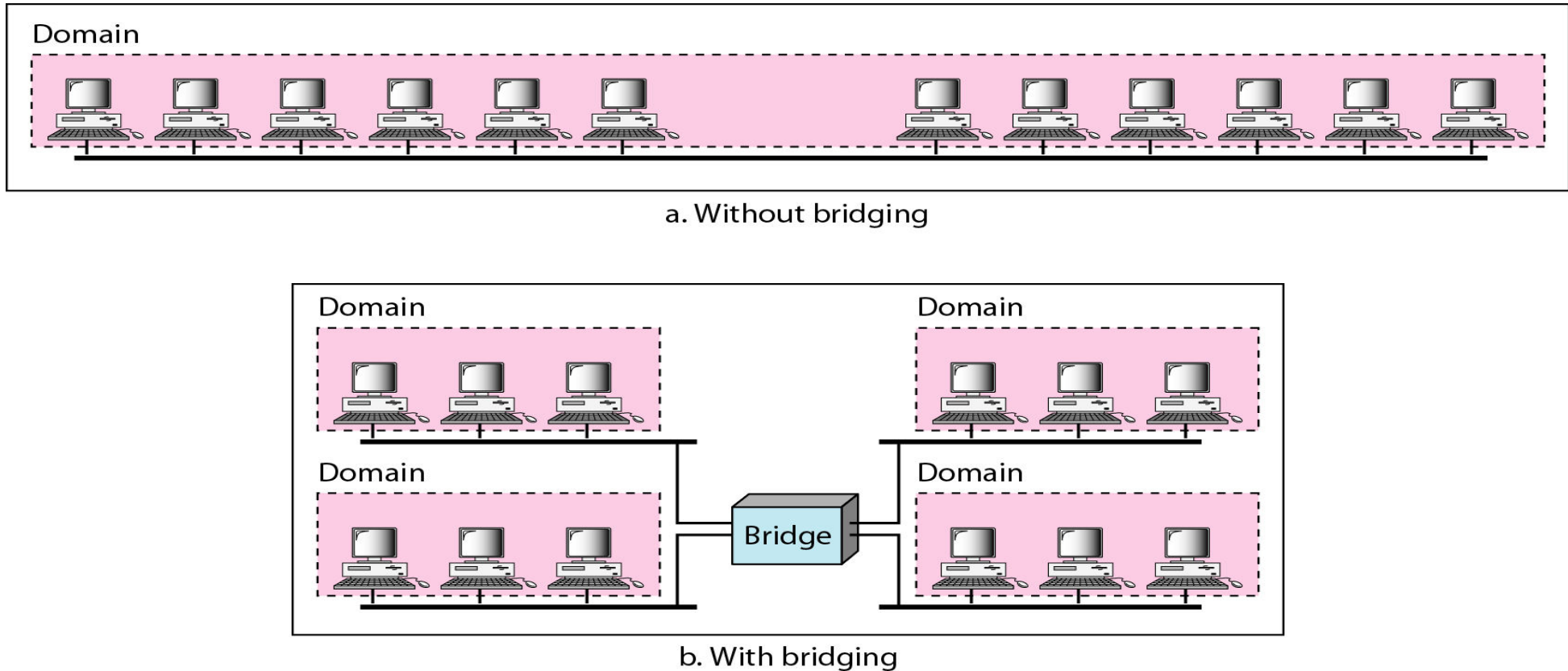
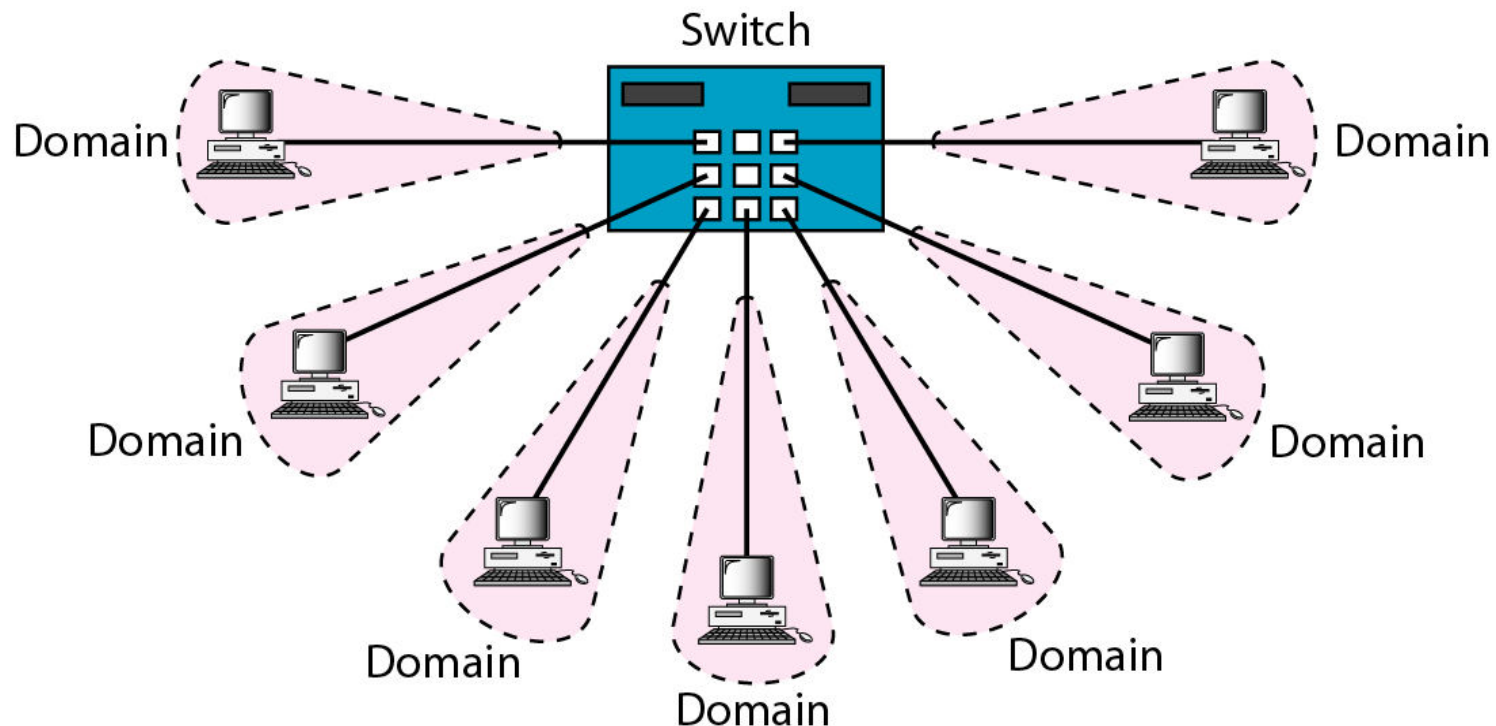


Figure 13.16 *Collision domains in an unbridged network and a bridged network*

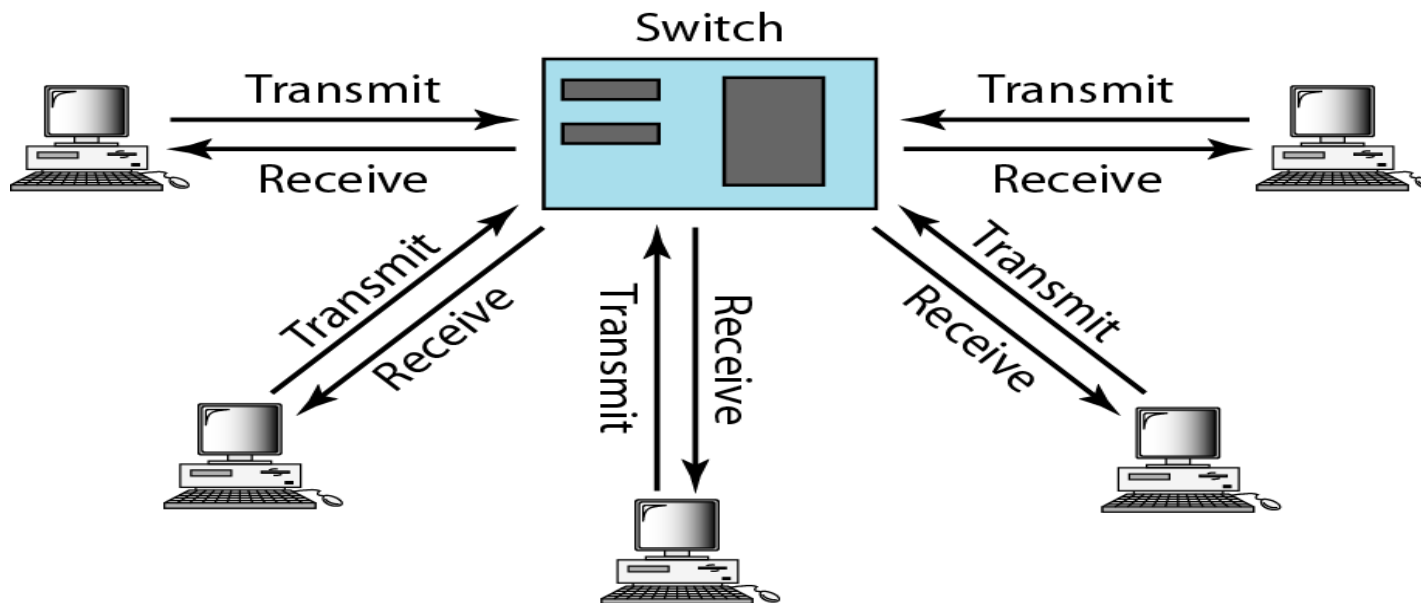
Switched Ethernet

- ❑ Bandwidth is shared only between the station and the switch (5 Mbps each)
- ❑ N-port switch; Switched Ethernet



Full-Duplex Ethernet

- ❑ A Limitation of 10Base5 and 10Base2 - half-duplex.
- ❑ Evolution : switched Ethernet → full duplex Switched Ethernet
 - ❖ 10Base-T is always Full-duplex.



- ❑ Full duplex mode increases the capacity of each domain from 10 to 20 Mbps.

Full-Duplex Ethernet

- ❑ **Traditional Ethernet was designed as a connectionless protocol at the MAC sublayer .**
 - ❖ **No explicit flow control or error control to inform the sender that the frame has arrived at the destination without error.**
- ❑ **For flow and error control in full-duplex switched Ethernet**
 - ❖ **No need for CSMA/CD, this functionality can be turned off.**
 - **Each link is a point-to-point dedicated path between the station and the switch.**
 - ❖ **Adding a sublayer called MAC Control between MAC sublayer and LLC sublayer**



13-4 FAST ETHERNET

Fast Ethernet was designed to compete with LAN protocols such as FDDI or Fiber Channel. IEEE created Fast Ethernet under the name 802.3u. Fast Ethernet is backward-compatible with Standard Ethernet, but it can transmit data 10 times faster at a rate of 100 Mbps.

Topics discussed in this section:

MAC Sublayer
Physical Layer



The goals of the Fast Ethernet

- ❑ Upgrade the data rate to 100Mbps.
- ❑ Make it compatible with Standard Ethernet.
- ❑ Keep the same 48 bit address.
- ❑ Use the same frame format.
- ❑ Keep the same minimum and maximum frame lengths.



MAC Sublayer - Fast Ethernet

- ❑ Evolution from 10 to 100 Mbps doesn't change the MAC sublayer.
- ❑ Access method is CSMA/CD
 - ❖ for the half-duplex approach, which is kept for backward compatibility;
 - ❖ for full duplex Fast Ethernet, there is no need for CSMA/CD
- ❑ Frame format, minimum and maximum frame lengths, and addressing are the same.

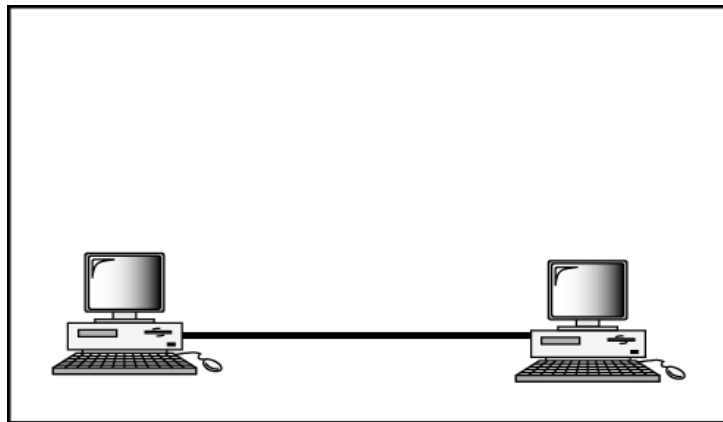


Autonegotiation

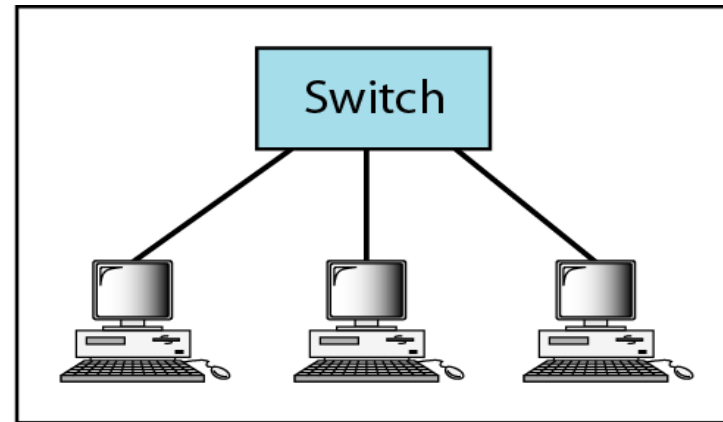
- ❑ **Allowing two devices to negotiate the mode or data rate of operation.**
 - ❖ **To allow incompatible devices to connect to one another. For example, between 10 Mbps-device and 100 Mbps-device**
 - ❖ **To allow one device to have multiple capabilities**
 - ❖ **To allow a station to check a hub's capabilities**



Physical layer – Fast Ethernet



a. Point-to-point



b. Star

Figure 13.19 *Fast Ethernet topology*

Physical layer – Fast Ethernet

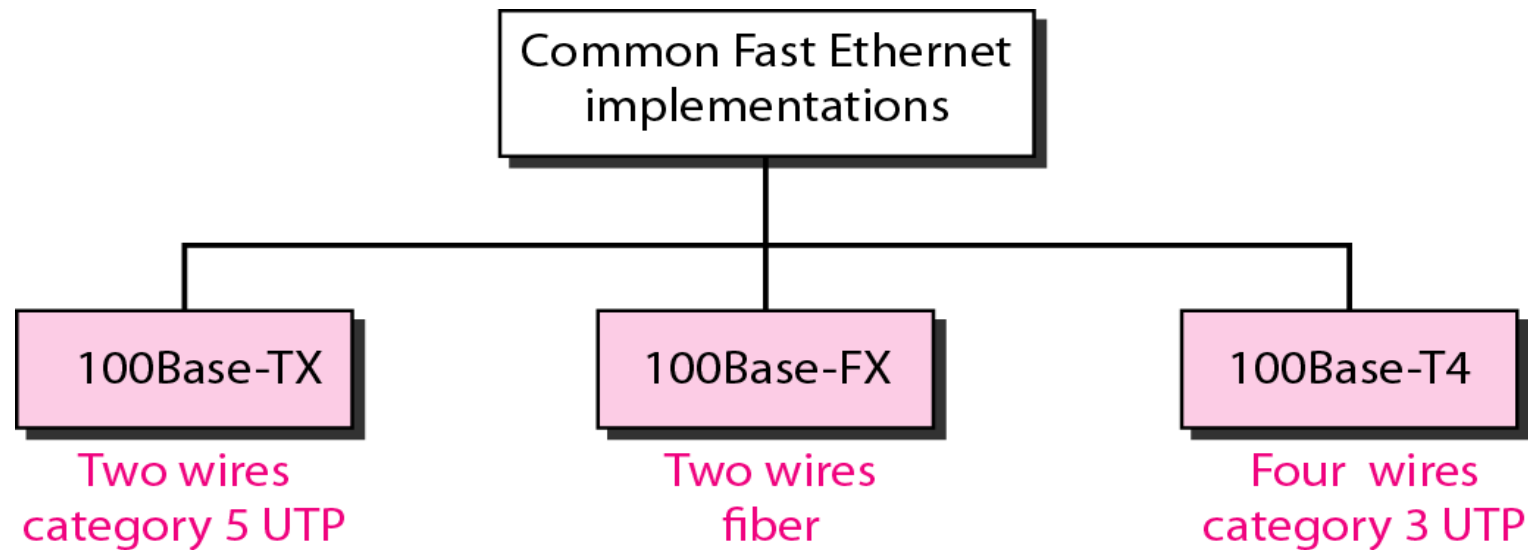
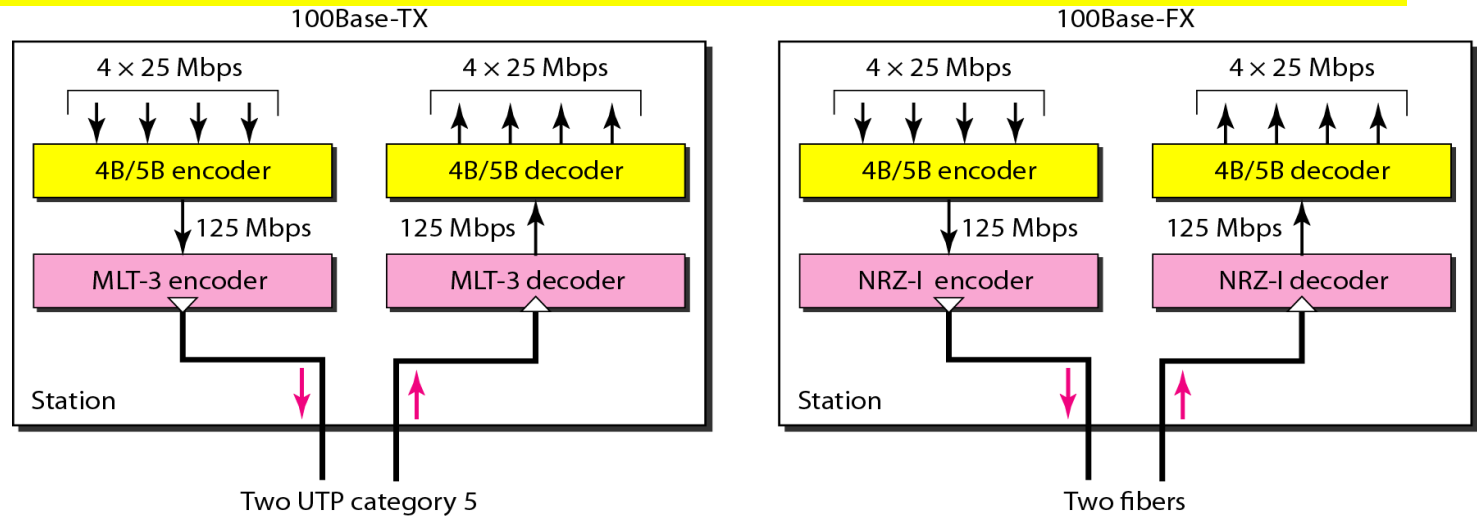


Figure 13.20 *Fast Ethernet implementations*



Physical layer – Fast Ethernet



□ Encoding/decoding – first performs block encoding using 4B/5B, then encoded using MLT-3 (multiline transmission, three level)

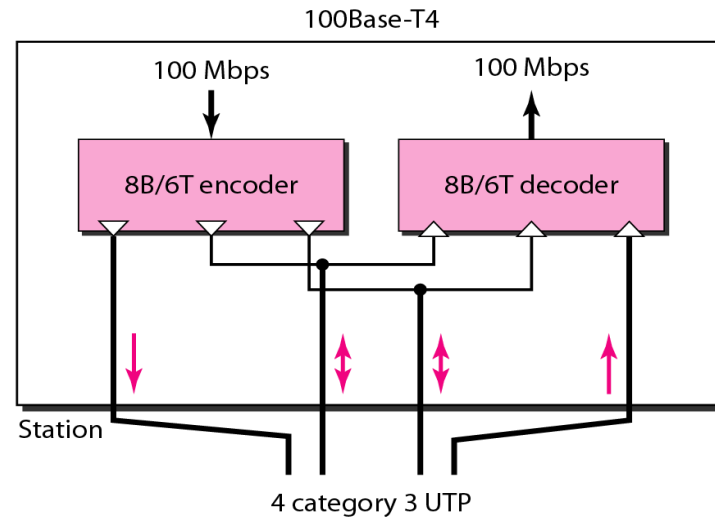


Figure 13.21 Encoding for Fast Ethernet implementation

Physical layer – Fast Ethernet

Table 13.2 *Summary of Fast Ethernet implementations*

<i>Characteristics</i>	<i>100Base-TX</i>	<i>100Base-FX</i>	<i>100Base-T4</i>
Media	Cat 5 UTP or STP	Fiber	Cat 4 UTP
Number of wires	2	2	4
Maximum length	100 m	100 m	100 m
Block encoding	4B/5B	4B/5B	
Line encoding	MLT-3	NRZ-I	8B/6T



13-5 GIGABIT ETHERNET

The need for an even higher data rate resulted in the design of the Gigabit Ethernet protocol (1000 Mbps). The IEEE committee calls the standard 802.3z.

Topics discussed in this section:

MAC Sublayer

Physical Layer

Ten-Gigabit Ethernet



The goals of the Gigabit Ethernet

- ❑ Upgrade the data rate to 1Gbps.
- ❑ Make it compatible with Standard or fast Ethernet.
- ❑ Use the same 48 bit address.
- ❑ Keep the same frame format.
- ❑ Keep the same minimum and maximum frame lengths.
- ❑ To support autonegotiation as defined in Fast Ethernet.



MAC Sulayer - Gigabit Ethernet

❑ Gigabit Ethernet has two approaches for Multiple Access

: half-duplex and full-duplex

❖ Almost all implementations of Gigabit Ethernet follow the full-duplex approach.

❖ Half-duplex approach to show that Gigabit Ethernet can be compatible with the previous generations (Standard & Fast Ethernet).

- ❖ In the full-duplex mode of Gigabit Ethernet, there is no collision;
- ❖ The maximum length of the cable is determined by the signal attenuation in the cable.



Physical Layer - Gigabit Ethernet

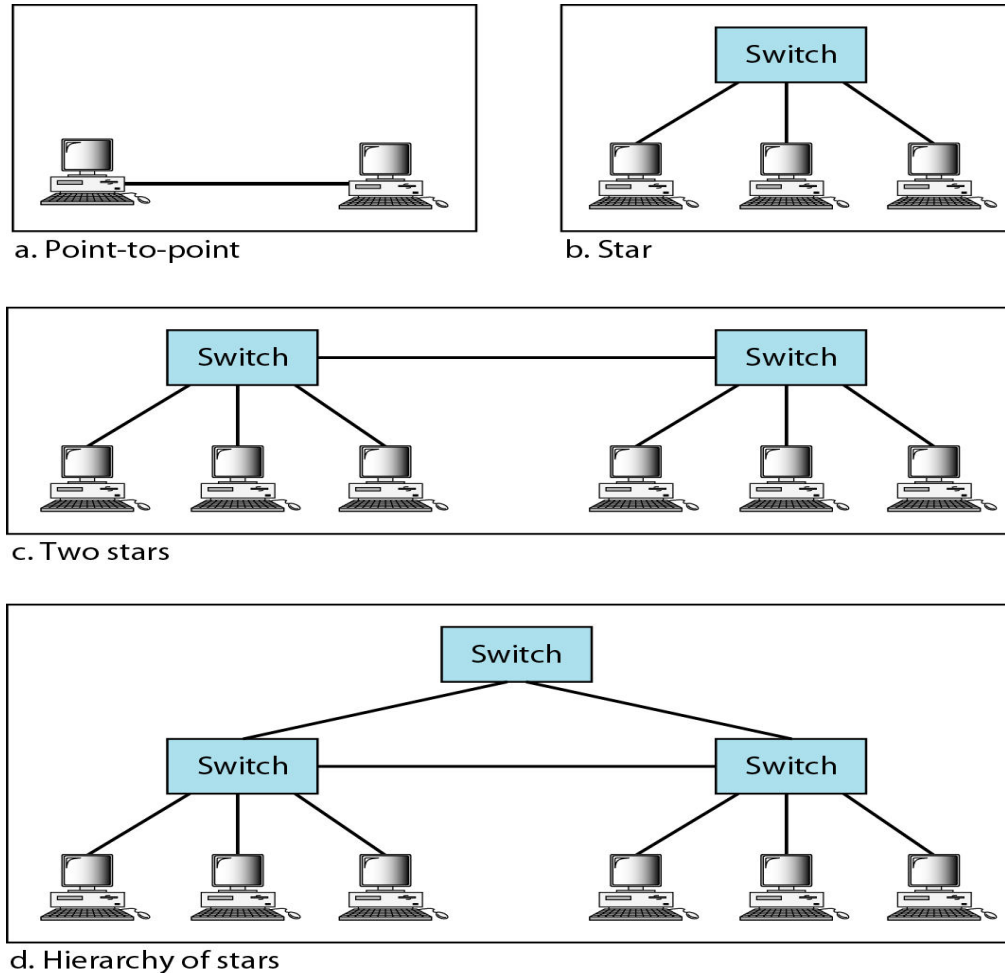


Figure 13.22 *Topologies of Gigabit Ethernet*



Physical Layer - Gigabit Ethernet

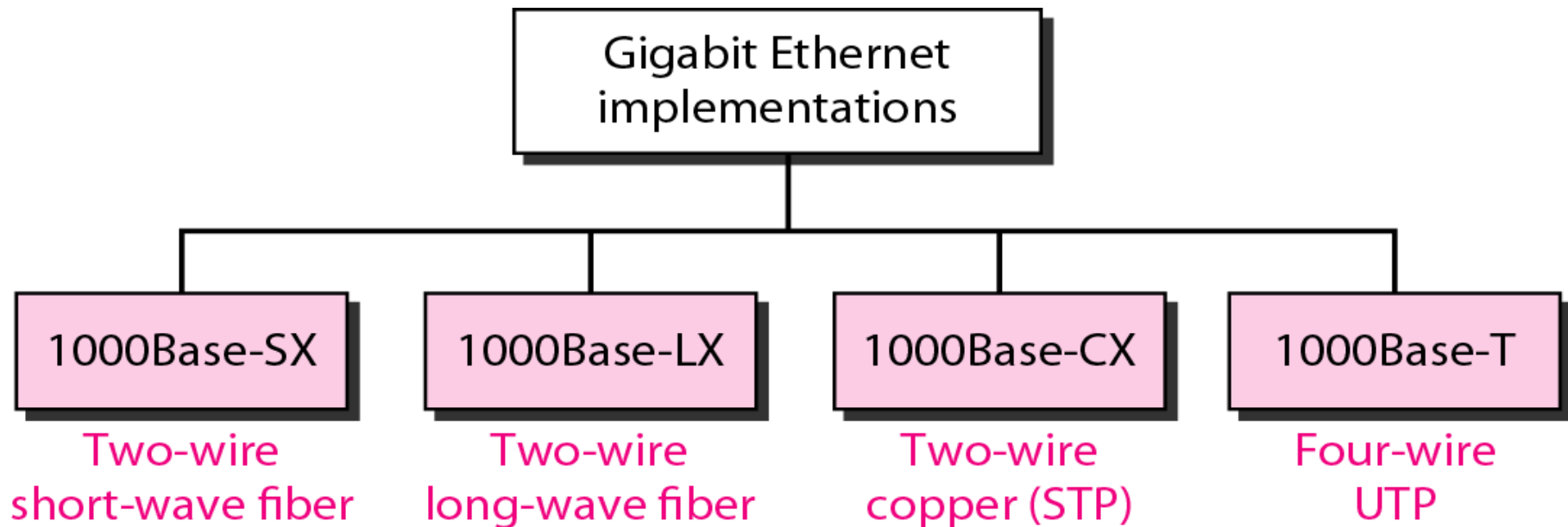


Figure 13.23 *Gigabit Ethernet implementations*

Physical Layer - Gigabit Ethernet

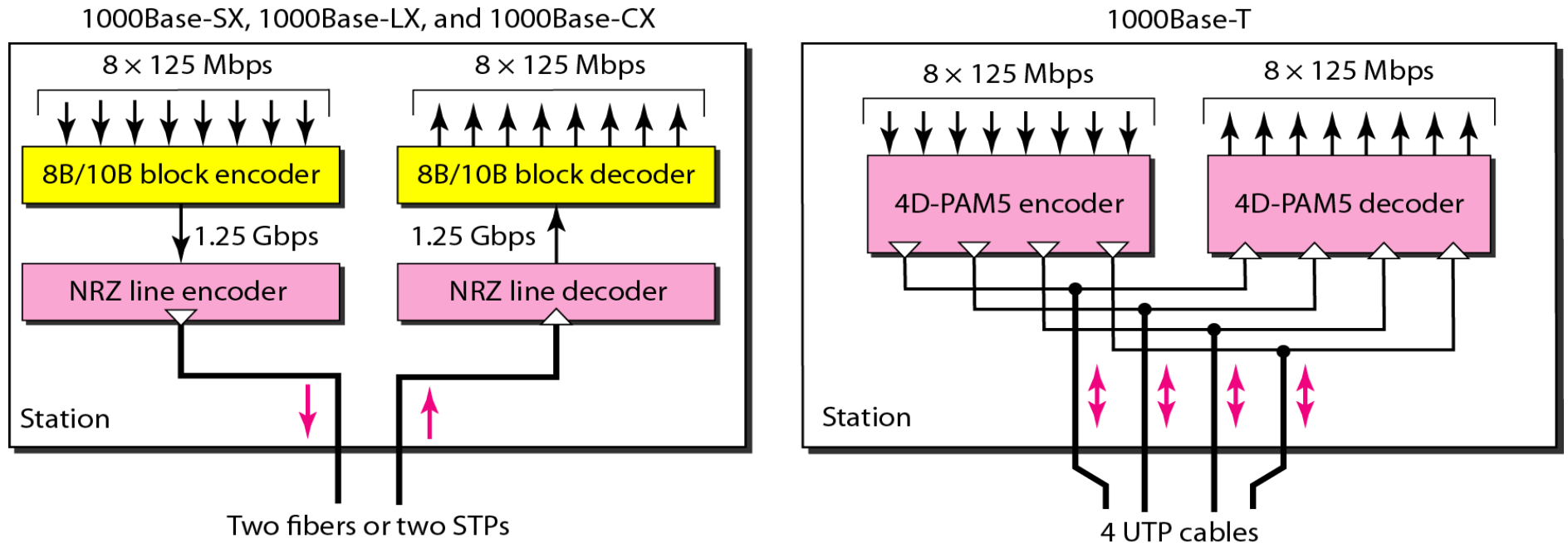


Figure 13.24 *Encoding in Gigabit Ethernet implementations*

Physical Layer - Gigabit Ethernet

Table 13.3 *Summary of Gigabit Ethernet implementations*

<i>Characteristics</i>	<i>1000Base-SX</i>	<i>1000Base-LX</i>	<i>1000Base-CX</i>	<i>1000Base-T</i>
Media	Fiber short-wave	Fiber long-wave	STP	Cat 5 UTP
Number of wires	2	2	2	4
Maximum length	550 m	5000 m	25 m	100 m
Block encoding	8B/10B	8B/10B	8B/10B	
Line encoding	NRZ	NRZ	NRZ	4D-PAM5



Physical Layer - Gigabit Ethernet

❑ MAC Sublayer

❖ Ten-Gigabit Ethernet operates only in full duplex mode which means there is no need for contention; CSMA/CD is not used in 10-Gigabit Ethernet.

❑ Physical Layer

❖ Physical layer is designed for using fiber-optic cable over long distances.

Table 13.4 *Summary of Ten-Gigabit Ethernet implementations*

<i>Characteristics</i>	<i>10GBase-S</i>	<i>10GBase-L</i>	<i>10GBase-E</i>
Media	Short-wave 850-nm multimode	Long-wave 1310-nm single mode	Extended 1550-nm single mode
Maximum length	300 m	10 km	40 km

Summary

- ❑ Ethernet is the most widely used local area network protocols.
- ❑ The IEEE 802.3 Standard defines 1-persistent CSMA/CD as the access method for first-generation 10-Mbps Ethernet.
- ❑ The data link layer of Ethernet consists of the LLC sublayer and the MAC sublayer.
- ❑ The MAC sublayer is responsible for the operation of the CSMA/CD access method and framing.
- ❑ Each station on an Ethernet network has a unique 48-bit address imprinted on its network interface card (NIC).
- ❑ The minimum frame length for 10-Mbps Ethernet is 64 bytes; the maximum is 1518 bytes.
- ❑ The common implementations of 10-Mbps Ethernet are 10Base5 (thick Ethernet), 10Base2 (thin Ethernet), 10Base-T (twisted-pair Ethernet), and 10Base-F (fiber Ethernet).



Summary (2)

- ❑ The 10Base5 implementation of Ethernet used thick coaxial cable. 10Base2 uses thin coaxial cables. 10Base-T uses four twisted-pair cables that connect each station to a common hub. 10Base-F used fiber-optic cable.
- ❑ A bridge can increase the bandwidth and separate the collision domains on an Ethernet LAN.
- ❑ A switch allows each station on an Ethernet LAN to have the entire capacity of the network to itself.
- ❑ Full-duplex mode doubles the capacity of each domain and removes the need for CSMA/CD method.
- ❑ Fast Ethernet has a data rate of 100 Mbps.



Summary (3)

- ❑ In Fast Ethernet, autonegotiation allows two devices to negotiate the mode or data rate of operation.
- ❑ The common Fast Ethernet implementations are 100Base-TX (two pairs of twisted pair cable), 100Base-FX (two fiber-optic cables), and 100Base-T4 (four pairs of voice-grade, or higher, twisted-pair cable).
- ❑ Gigabit Ethernet has a data rate of 1000Mbps.
- ❑ Gigabit Ethernet access methods include half-duplex mode using traditional CSMA/CD (not common) and full-duplex mode (most popular method).

Summary (4)

- ❑ The common Gigabit Ethernet implementations are 1000Base-SX (two optical fibers and a short-wave laser source), 1000Base-LX (two optical fibers and a long-wave laser source), and 1000Base-T (four twisted pairs).
- ❑ The latest Ethernet standard is Ten-Gigabit Ethernet that operates at 10 Gbps. The three common implementations are 10GBase-S, 10GBase-L, and 10GBase-E. These implementations use fiber-optic cables in full-duplex mode.